INSURANCE PORTFOLIO ANALYTICS REPORT

Enterprise Insurance Performance & Risk Assessment

1. EXECUTIVE SUMMARY

This report evaluates the performance, risk exposure, and operational efficiency of a corporate insurance portfolio using a modern data warehouse architecture.

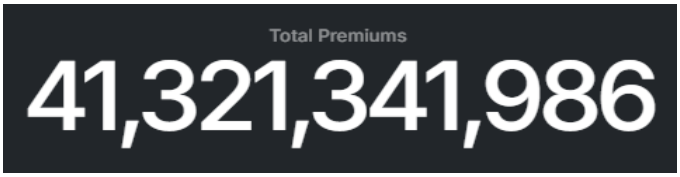
The analysis integrates policy, claims, and employee-level data to provide **end-to-end visibility into profitability drivers, risk concentrations, and operational bottlenecks**.

Key Stakeholder Questions Answered

- Is the insurance portfolio profitable?
- Which policy types are driving losses?
- Where is risk concentrated (coverage, employees)?
- How efficient is claims processing?
- Which insurers and products perform best?

2. PORTFOLIO OVERVIEW

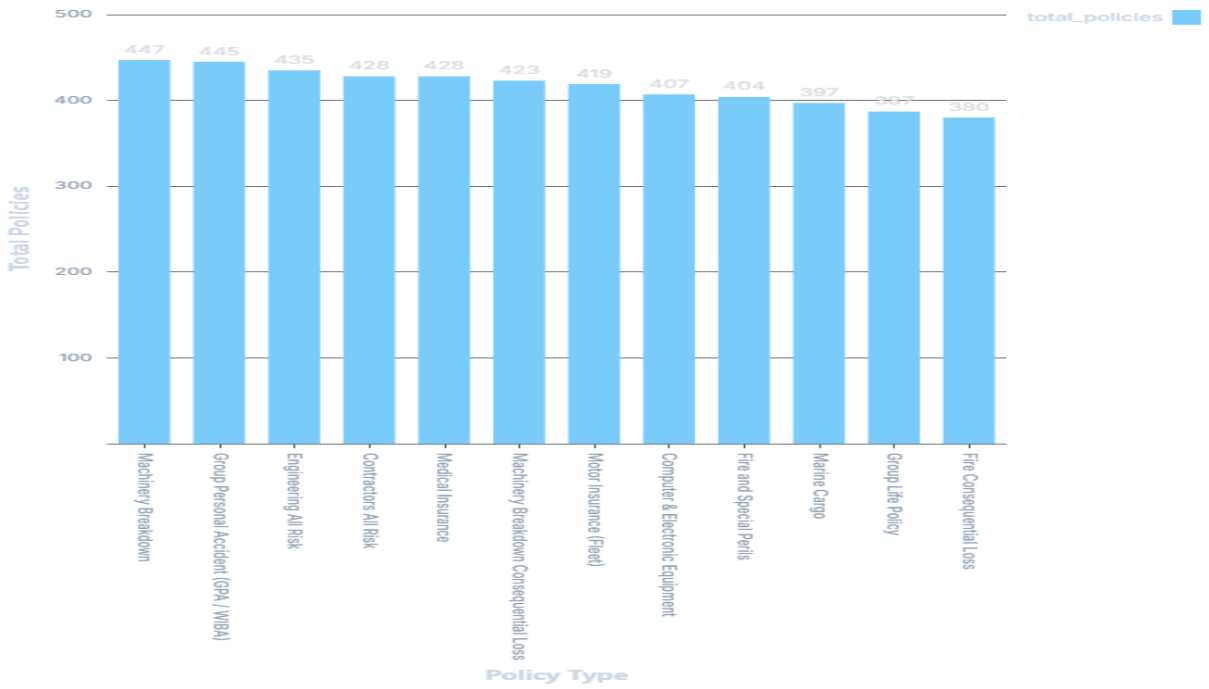
2.1 Key KPIs



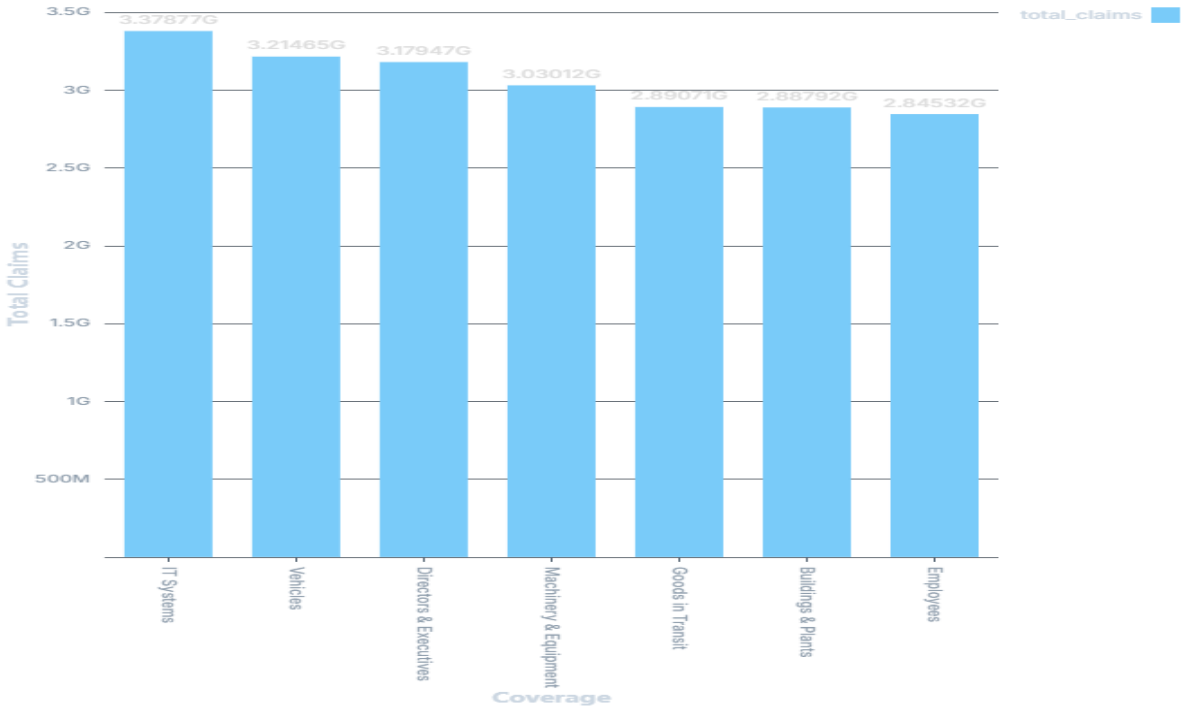
	total_claims	total_premiums	loss_ratio		total_claims	total_policies	claim_frequency
1	21426960759.00	66565718790.00	0.32	1	8000	3988	2.01

2.2 Policy Distribution

- Portfolio is diversified across multiple insurance products
- However, **volume ≠ profitability**. High-volume policies may still be loss-making



2.3 Coverage Distribution



- Coverage types tied to **physical assets (machinery, buildings)** tend to carry higher financial exposure
- Employee-based policies typically have **high frequency but lower severity**

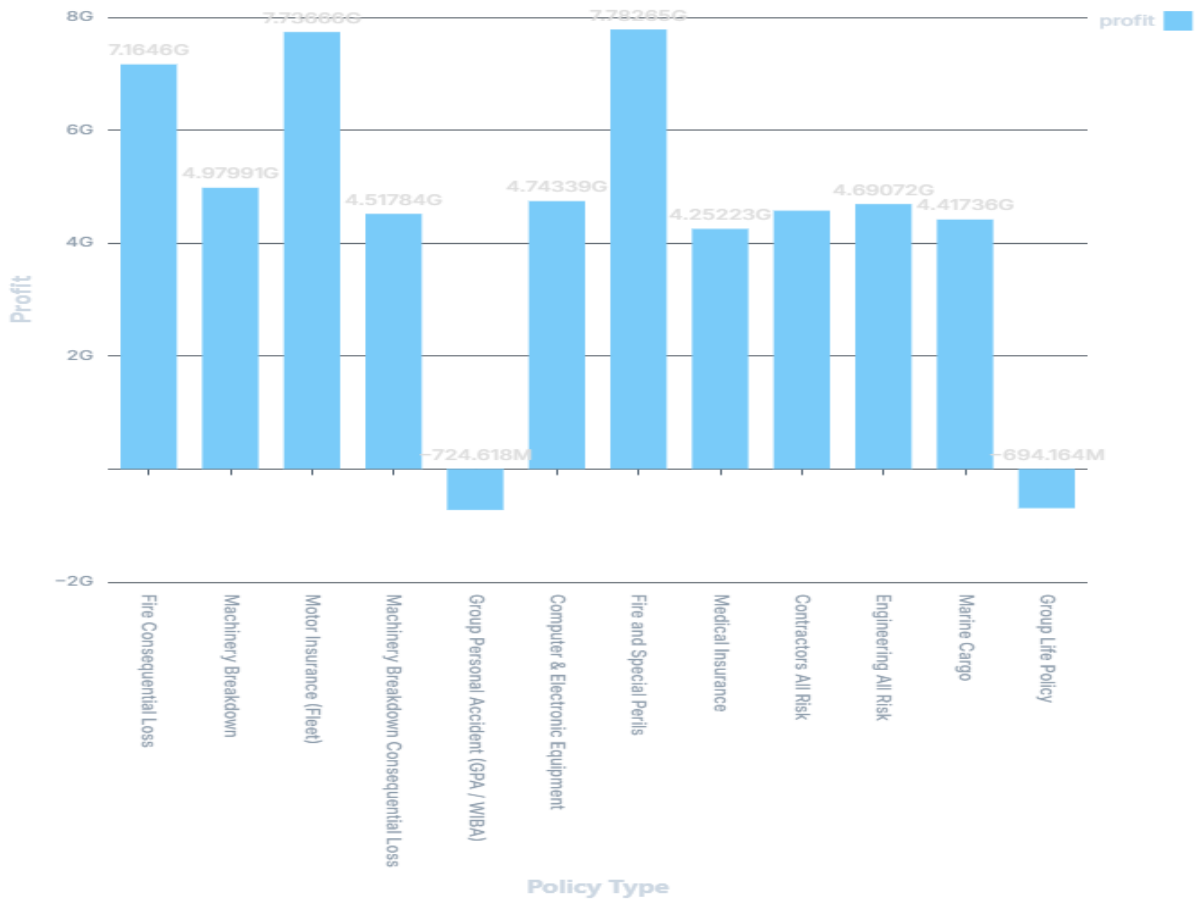
3. FINANCIAL PERFORMANCE ANALYSIS

3.1 Premiums vs Claims

	total_claims	total_premiums	loss_ratio
1	21426960759.00	66565718790.00	0.32

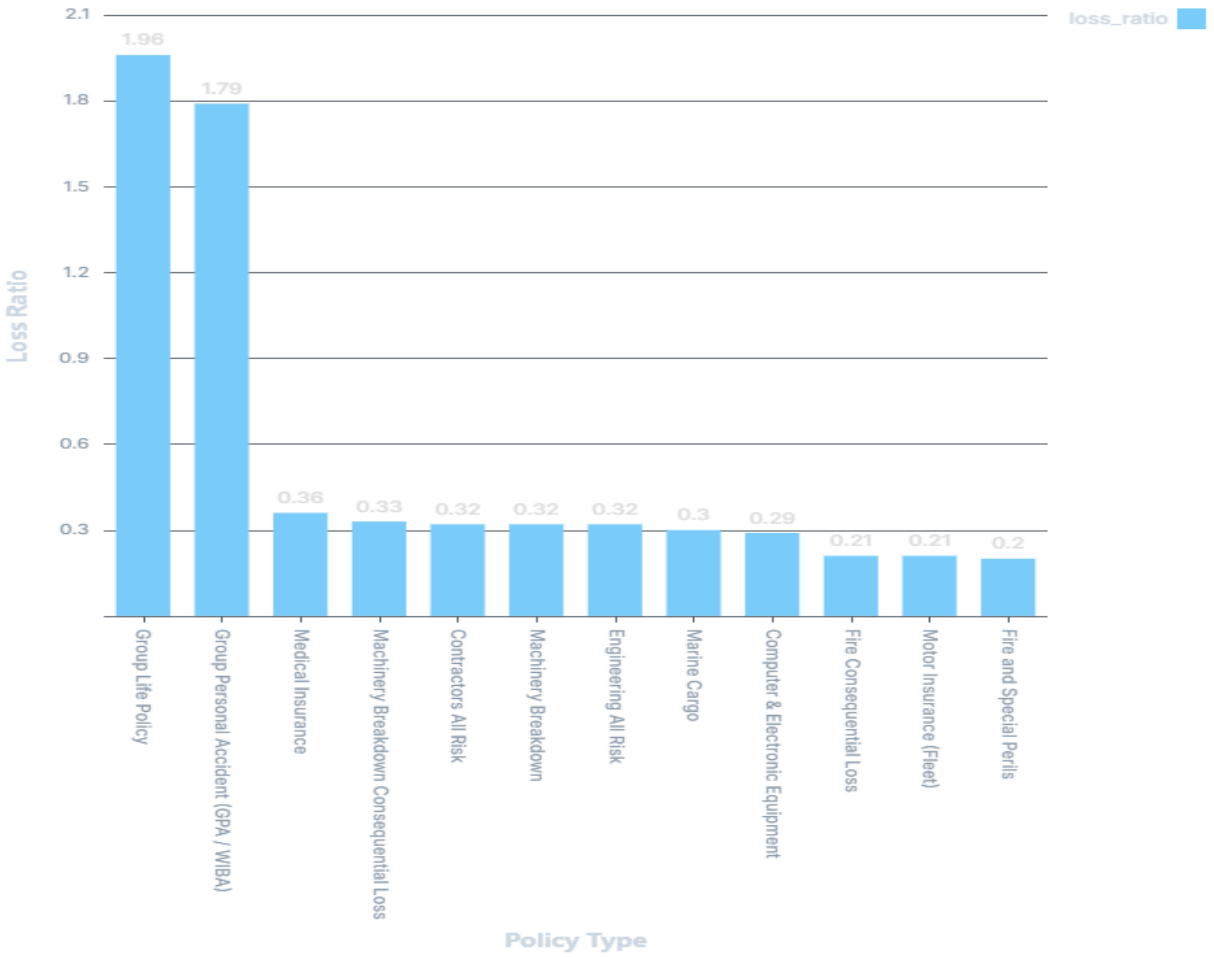
- The gap between premiums and claims determines sustainability
- Narrow gaps signal **pricing or underwriting issues**

3.2 Policy Profitability



- Profitability varies significantly across products
- Some policies:
    - Generate strong premiums but low claims - **high-value**
    - Generate high claims - **portfolio drag**

3.3 Loss Ratio Analysis



- **> 1.0 -** High claims recovery - High risk exposure (adverse events occurred)
- **0.7 - 1.0 -** Balanced - Efficient risk transfer (optimal zone)
- **< 0.7 -** Low claims - Low utilization / high insurance cost

Certain policy types consistently exceed acceptable thresholds - **pricing misalignment or high-risk exposure**.

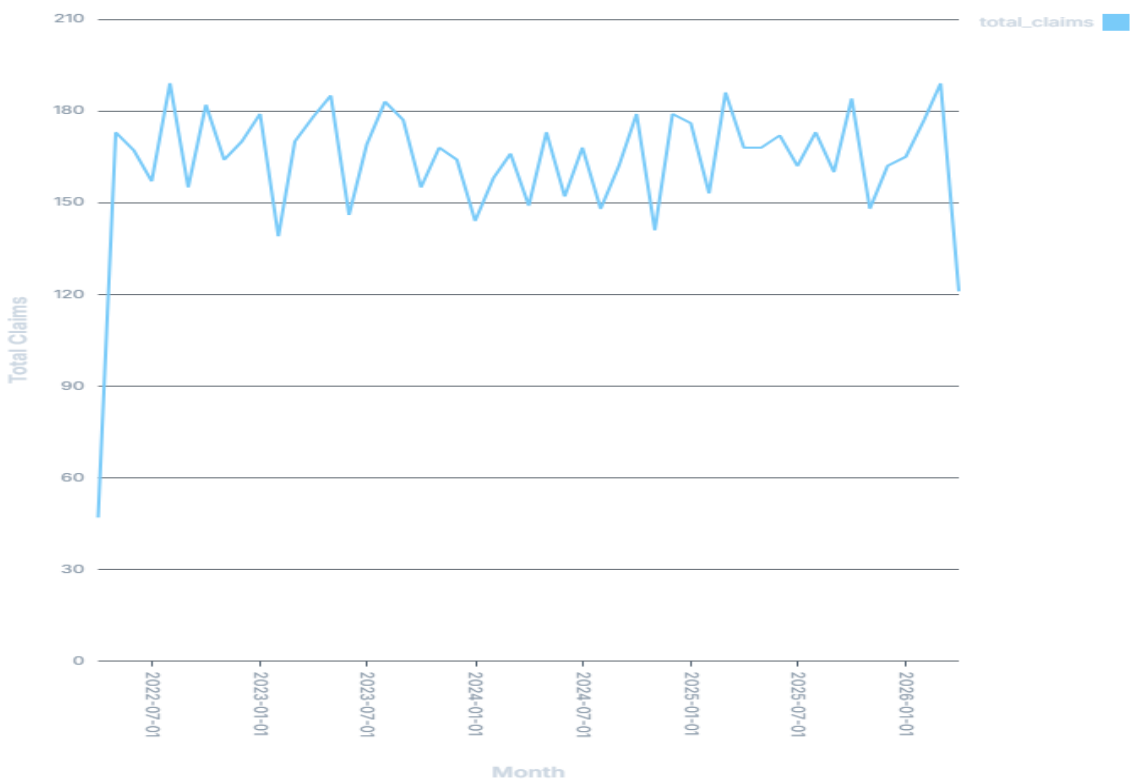
4. CLAIMS & RISK ANALYSIS

4.1 Claim Frequency

	total_claims	total_policies	claim_frequency
1	8000	3988	2.01

- High frequency = operational risk or weak controls
- Low frequency but high severity = catastrophic risk exposure

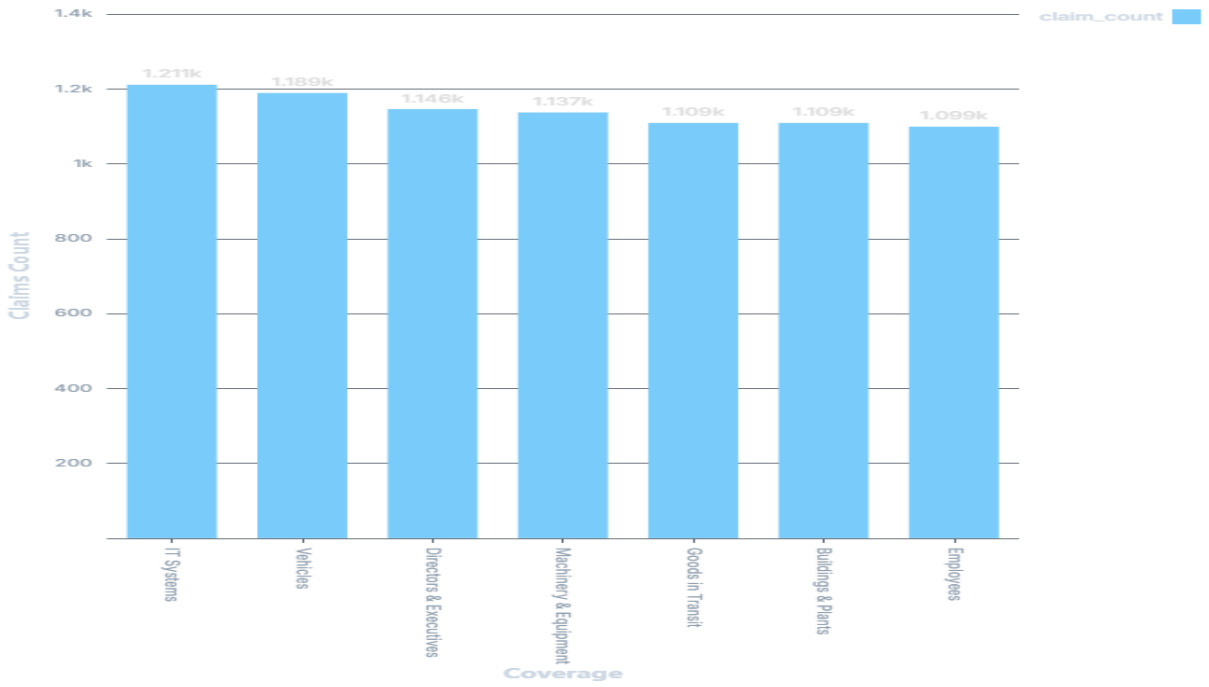
4.2 Claims Trend Over Time



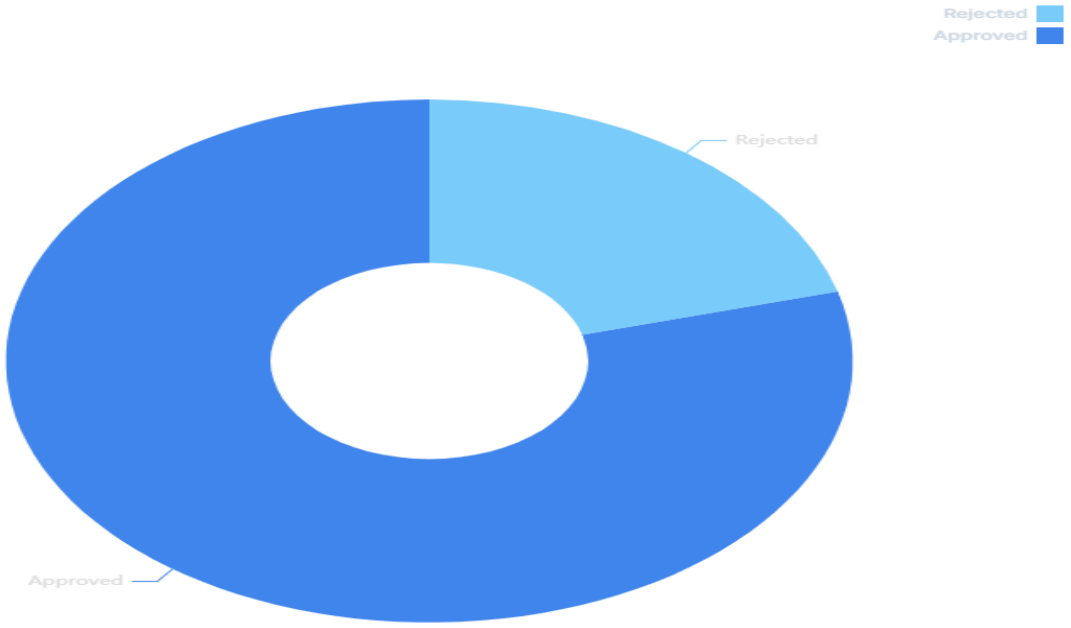
- Presence of seasonality patterns and claim spikes
- Indicates potential operational or environmental risks

4.3 Risk Exposure by Coverage

- A small number of coverage types typically drive majority of claims (Pareto effect)
- Critical for **risk mitigation strategy**



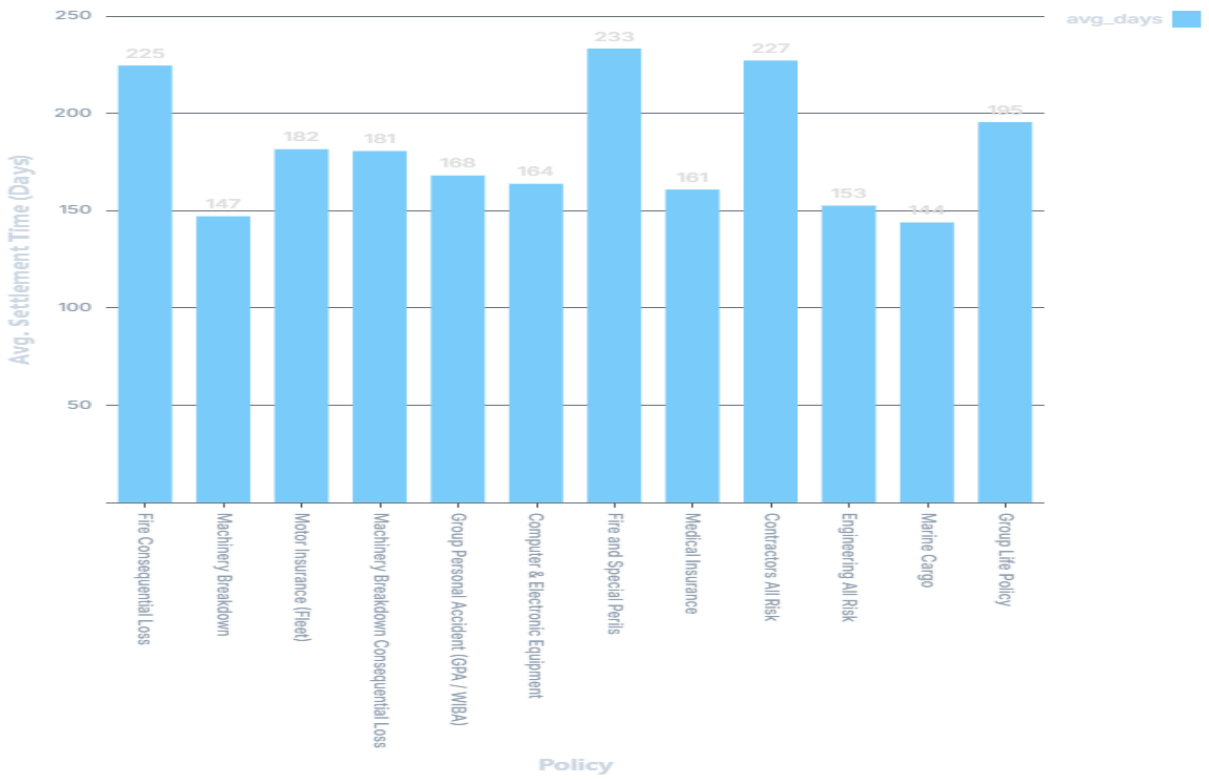
4.4 Claim Status Analysis



- High rejection rates may indicate strong fraud detection or poor underwriting / unclear policy terms

5. OPERATIONAL EFFICIENCY

5.1 Settlement by Policy Type



- Complex policies (engineering, fire) take longer
- Simpler policies should be optimized for automation

6. EMPLOYEE & ORGANIZATIONAL RISK

6.1 Employee Risk

	employee_type	total_claims	total_paid	avg_approval_ratio
1	Manager	1993	5326690440.00	0.59909683893627696939
2	Ex-Staff	2006	5468224956.00	0.60119641076769690927
3	Staff	1964	5189875916.00	0.57841140529531568228
4	Director	2037	5442169447.00	0.59695630829651448208

- Different employee groups show different risk behavior:
    - Staff - high volume
    - Executives - lower frequency, higher severity

7. INSURER PERFORMANCE

	insurer_name	total_claims	total_paid	avg_claim_size
1	APA Insurance	1388	3792829843.00	2732586
2	Britam	1422	3680647938.00	2588360
3	ICEA Lion	1289	3586843212.00	2782656
4	Old Mutual	1327	3563489177.00	2685372
5	Jubilee Insurance	1334	3523517852.00	2641318
6	CIC Insurance	1240	3279632737.00	2644865

- Some insurers pay faster, handle higher volumes and or manage claim sizes differently.
- Useful for **renewal negotiations & vendor strategy**

8. KEY INSIGHTS

Financial

- Profitability is uneven - driven by specific policy types
- Loss ratio highlights underpriced or high-risk segments

Risk

- Risk is concentrated in asset-heavy coverage (machinery, buildings) and specific policy categories

Operational

- Settlement times vary widely - inefficiencies
- Claims process is not standardized across policy types

Organizational

- Risk is linked to employee category and department
- Not all groups contribute equally to claims

9. STRATEGIC RECOMMENDATIONS

9.1 Pricing & Underwriting

- Reprice high loss-ratio policies
- Tighten underwriting for engineering and Fire-related coverage
- Introduce risk-based pricing models

9.2 Risk Management

- Reduce exposure to high-loss coverage categories
- Introduce preventive maintenance programs (for machinery risk)
- Strengthen internal controls for high-frequency claims

9.3 Operational Efficiency

- Automate low-complexity claims (motor, employee claims)
- Set SLA targets for claim settlement
- Introduce workflow tracking for bottlenecks

9.4 Portfolio Optimization

- Scale profitable policy types
- Reduce reliance on high-risk segments
- Diversify coverage strategically

9.5 Insurer Strategy

- Benchmark insurers on:
    - Claim payout efficiency
    - Settlement time
    - Claim severity handling
- Use insights for contract renegotiation